

Motion Alarm Project

Arduino Uno + PIR Motion Sensor + Buzzer + LED

Project Overview

Yeh project ek PIR (Passive Infrared) motion sensor ka istemal karke movement detect karta hai. Jab motion detect hoti hai, PIR sensor Arduino ke digital pin 2 par HIGH signal bhejta hai. Arduino phir buzzer aur red LED ko automatically ON kar deta hai. Jab movement nahi hoti, dono OFF rehte hain. Is project mein decision simple if/else condition se hota hai.

Components Used

#	Component	Quantity	Purpose
1	Arduino Uno R3	1	Main microcontroller board
2	PIR Motion Sensor (HC-SR501)	1	Detects movement / human motion
3	Buzzer (piezo buzzer)	1	Produces warning alarm sound
4	LED (red)	1	Visual warning indicator
5	Resistor - 220 ohm	1	Current-limiting resistor for LED
6	Breadboard	1	To hold and connect components
7	Jumper wires	as needed	Connections between components

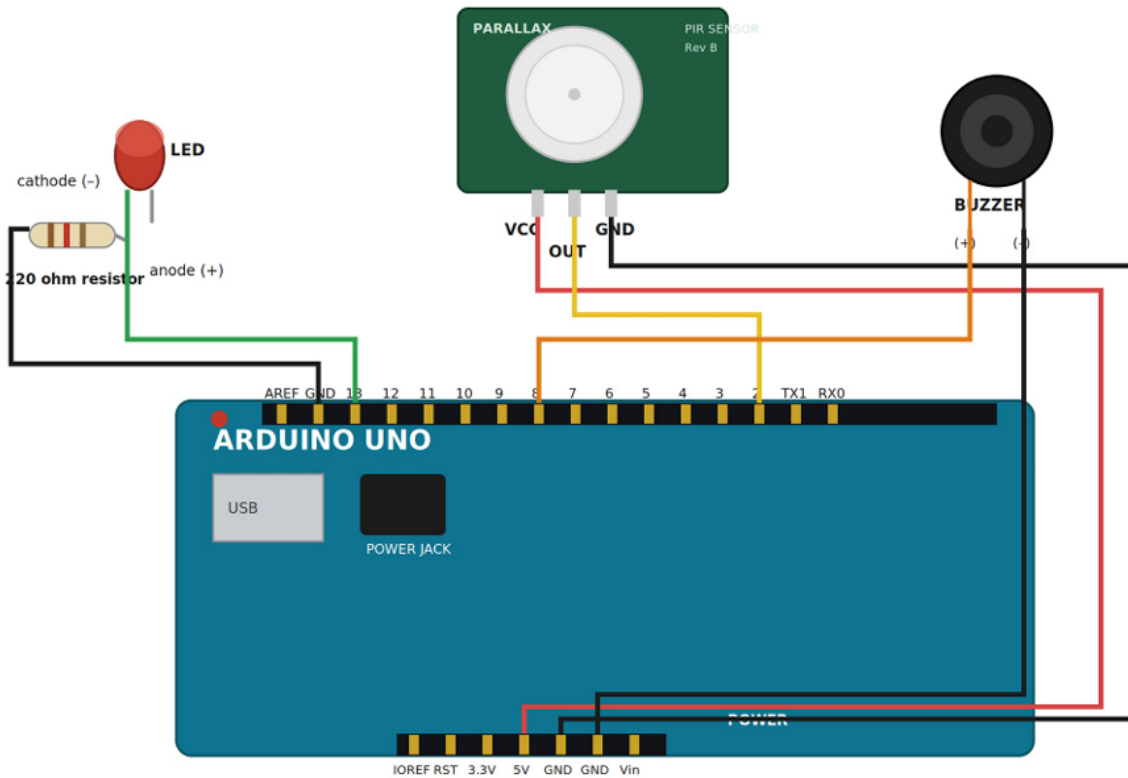
Build Steps

1	Tinkercad kholo - Naya circuit banao aur components search karo: "PIR Motion Sensor", "Buzzer", "LED", "Resistor".
2	PIR Sensor connect karo - VCC → Arduino 5V, GND → Arduino GND, OUT → digital pin 2.
3	Buzzer connect karo - Positive (+) → Arduino pin 8, Negative (-) → GND.
4	LED connect karo - Anode (lambi leg) → pin 13; cathode (choti leg) → 220Ω resistor → GND.
5	Code likho - Neeche di gayi Arduino sketch use karo.
6	Simulation run karo - "Start Simulation" click karo. PIR ke saamne mouse/cursor move karke motion trigger karo.

Circuit Diagram

Arduino Uno + PIR Motion Sensor + Buzzer + LED

Circuit Diagram



- PIR VCC → 5V
- PIR OUT → Pin 2
- GND connections
- Buzzer (+) → Pin 8
- LED anode → Pin 13

Full Arduino Code

Motion Alarm Project • PIR Sensor + Buzzer + LED

```
// Motion Alarm Project
// PIR Sensor + Buzzer + LED

int pirPin = 2;          // PIR sensor output pin
int buzzerPin = 8;       // Buzzer pin
int ledPin = 13;         // LED pin
int motionState = LOW;   // current motion status

void setup()
{
  pinMode(pirPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  motionState = digitalRead(pirPin);

  if (motionState == HIGH)
  {
    // Motion detected -> alarm on
    digitalWrite(buzzerPin, HIGH);
    digitalWrite(ledPin, HIGH);
    Serial.println("Motion Detected! Alarm ON");
  }
  else
  {
    // No motion -> alarm off
    digitalWrite(buzzerPin, LOW);
    digitalWrite(ledPin, LOW);
    Serial.println("No Motion");
  }

  delay(500); // readings stable rakhne ke liye
}
```

How the Project Works

1	Motion Detection: PIR sensor human/movement detect karta hai aur pin 2 par HIGH signal bhejta hai.
2	Arduino Decision: Arduino digitalRead(pirPin) se sensor ki state read karta hai.
3	Alarm ON: Agar state HIGH ho, Arduino buzzer aur LED ko HIGH karke ON kar deta hai.
4	Alarm OFF: Agar state LOW ho, Arduino buzzer aur LED ko LOW karke OFF rakhta hai.
5	Serial Monitor: Arduino status print karta hai: "Motion Detected! Alarm ON" ya "No Motion".

Wiring Summary

Component	Pin	Arduino Pin
PIR Sensor	VCC	5V
PIR Sensor	OUT	Digital Pin 2

Component	Pin	Arduino Pin
PIR Sensor	GND	GND
Buzzer	Positive (+)	Digital Pin 8
Buzzer	Negative (-)	GND
LED	Anode (+)	Digital Pin 13
LED	Cathode (-)	Via 220Ω resistor → GND

Tinkercad Circuit Screenshot

